Deep Learning and Deterministic Algorithms for Document Verification

Authors: Dr. Alan Turing, Dr. Ada Lovelace | Document Registration ID: ARXIV-2024-8841 | Published: October 15, 2024

1. Abstract

We present a robust verification engine pairing DP token alignment with visual differencing.
Across benchmark tests on multi-source documents, our model yields an empirical accuracy of 98.4%.

2. Introduction

Document tampering represents a substantial threat across academic admissions and audit workflows.
Traditional text-only extraction frequently ignores micro-typographic shifts, font changes, and layout variance.

3. Methodology

Our architecture combines the open-source pdf-diff bounding box tracker with PyMuPDF layout analysis.
Computer vision differencing with OpenCV captures altered diagram contours and embedded imagery.

4. Results

Rigorous empirical evaluation demonstrates measured precision of 98.4% with sub-second latency.

5. Conclusion

The integrated pipeline effectively pinpoints both typographic variance and visual document modifications.